

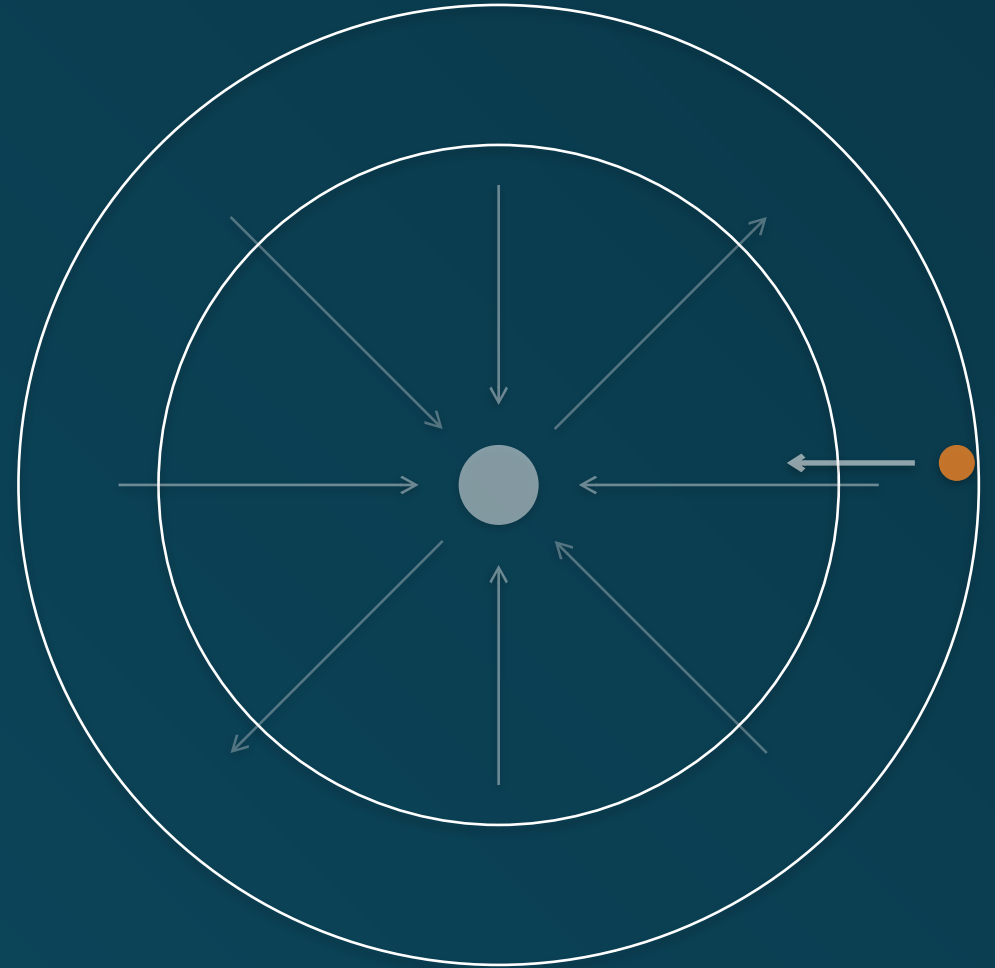
TOPIC 13 · A2 PHYSICS

GRAVITATIONAL FIELDS

引力场

Field strength · Newton's law · orbits · potential

场强 · 万有引力定律 · 圆轨道 · 引力势



Learning Outcomes

学习目标 — 完成本主题后你应掌握的内容

LEARNING OUTCOMES

学习目标

By the end of this topic, you will be able to:

完成本主题学习后，你应能够：

13.1 Gravitational field 引力场

- Understand a gravitational field as a field of force; define it as force per unit mass
- Represent a gravitational field by means of field lines 用场线表示引力场

13.2 Gravitational force between point masses

- Treat a uniform sphere as a point mass at its centre (from outside)
- Recall and use Newton's law of gravitation
- Analyse circular orbits by linking gravitational and centripetal force
- Understand the geostationary orbit 地球同步轨道 (24 h, equatorial, west to east)

13.3 Gravitational field of a point mass 点质量的场

- Derive from Newton's law and the definition of field
- Recall and use
- Explain why g is approximately constant near the Earth's surface

13.4 Gravitational potential 引力势

- Define potential as work done per unit mass from infinity to the point
- Use for the field of a point mass
- Understand potential energy of two point masses and use

The Gravitational Field

引力场的概念 — 力场、场强定义与场线表示

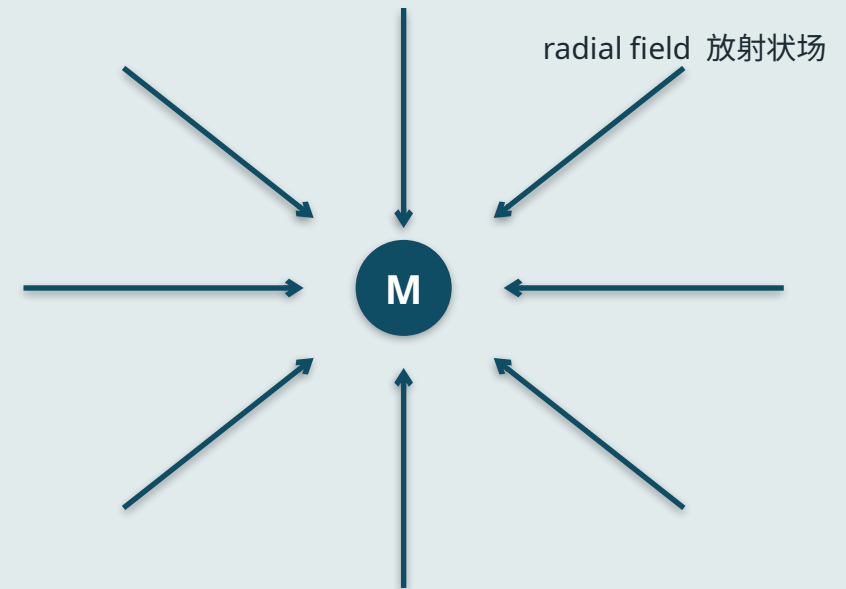
A gravitational field is a region of space where a mass experiences a force. Every mass attracts every other mass.

引力场：质量在其中会受到力的空间区域；一切质量互相吸引。

$$g = \frac{F}{m}$$

field strength = force per unit mass
unit: N kg^{-1} ($= \text{m s}^{-2}$) 单位

- **Field lines 场线**: the arrow shows the direction of the force on a point mass 箭头即受力方向
- Lines are smooth curves that never touch or cross 场线永不相交
- The closer the lines, the stronger the field 场线越密，场越强
- Field strength g is a **vector** 矢量, equal to the acceleration of free fall



Field of a point mass: lines converge on the centre 场线汇聚于质心

Newton's Law of Gravitation

万有引力定律 — 平方反比的吸引力

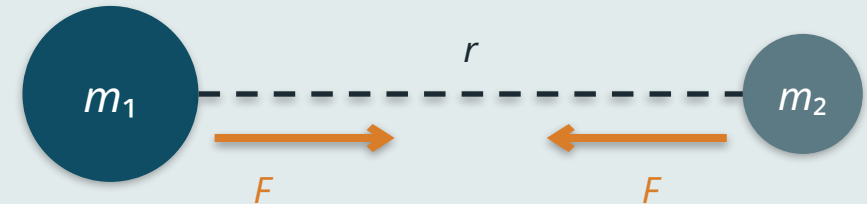
Two point masses attract each other with a force proportional to the **product of their masses** and inversely proportional to the **square of their separation**.

两质点间的引力与质量乘积成正比，与间距平方成反比。

$$F = \frac{G m_1 m_2}{r^2}$$

$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
 r = separation of centres 质心间距

- Valid for **point masses**, or spheres whose size $\ll$ separation
尺寸远小于间距时可视为质点
- A uniform sphere acts as a point mass at its centre 均匀球体等效于质心处质点
- Inverse-square 平方反比**: doubling r quarters the force
- Always attractive; equal and opposite on both masses 始终为吸引力



Equal and opposite attractive forces 等大反向的吸引力, r = centre-to-centre

Field of a Point Mass: $g = GM/r^2$

点质量的引力场强 — 由定义与万有引力定律推导

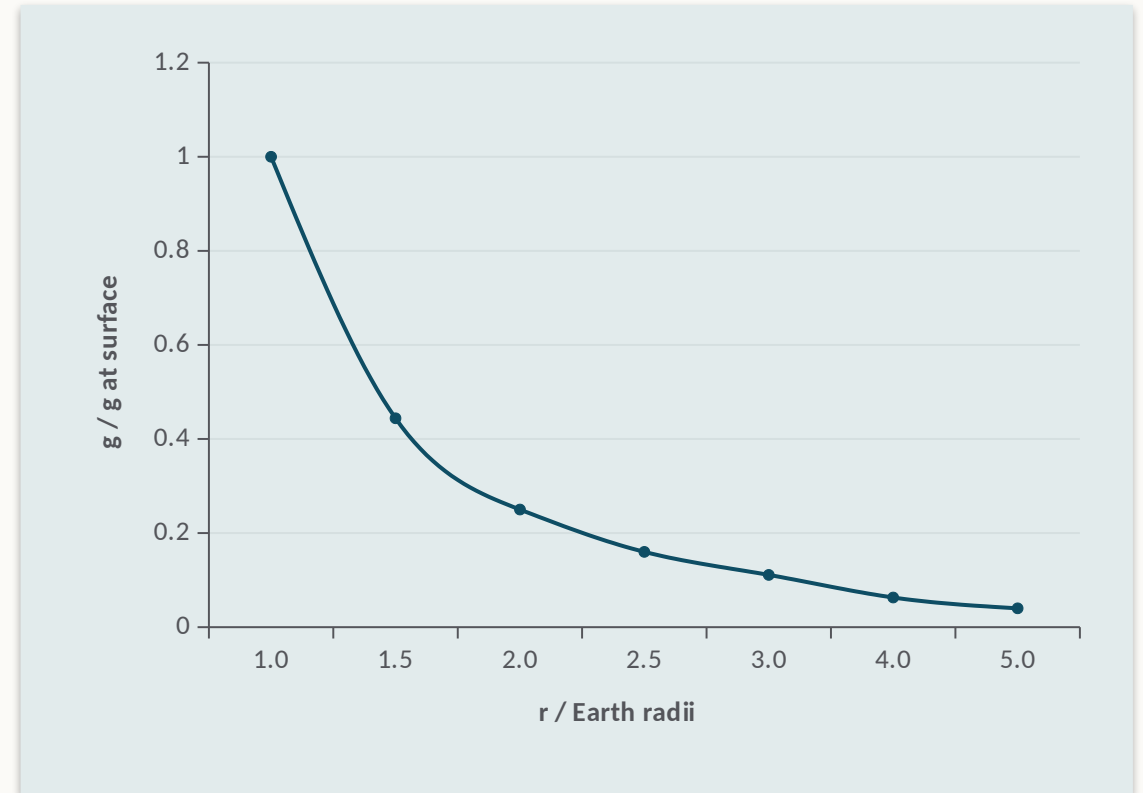
Derivation 推导 — force on a test mass m near mass M :

and

$$g = \frac{GM}{r^2}$$

M = mass of the central body
 r = distance from its centre 距质心

- At the Earth's surface:
- g follows an **inverse-square law**: at $r = 2R$ it falls to $g/4$ 平方反比衰减
- Near the surface, height changes are tiny vs 6.38×10^6 m, so g is **approximately constant** 地表附近 g 近似不变



$g \propto 1/r^2$ — field strength falls off rapidly with distance 随距离平方衰减

Circular Orbits

圆轨道分析 — 引力提供向心力

The **gravitational force provides the centripetal force** for a satellite of mass m orbiting a central mass M :

引力 = 向心力

$$v = \sqrt{\frac{GM}{r}}$$

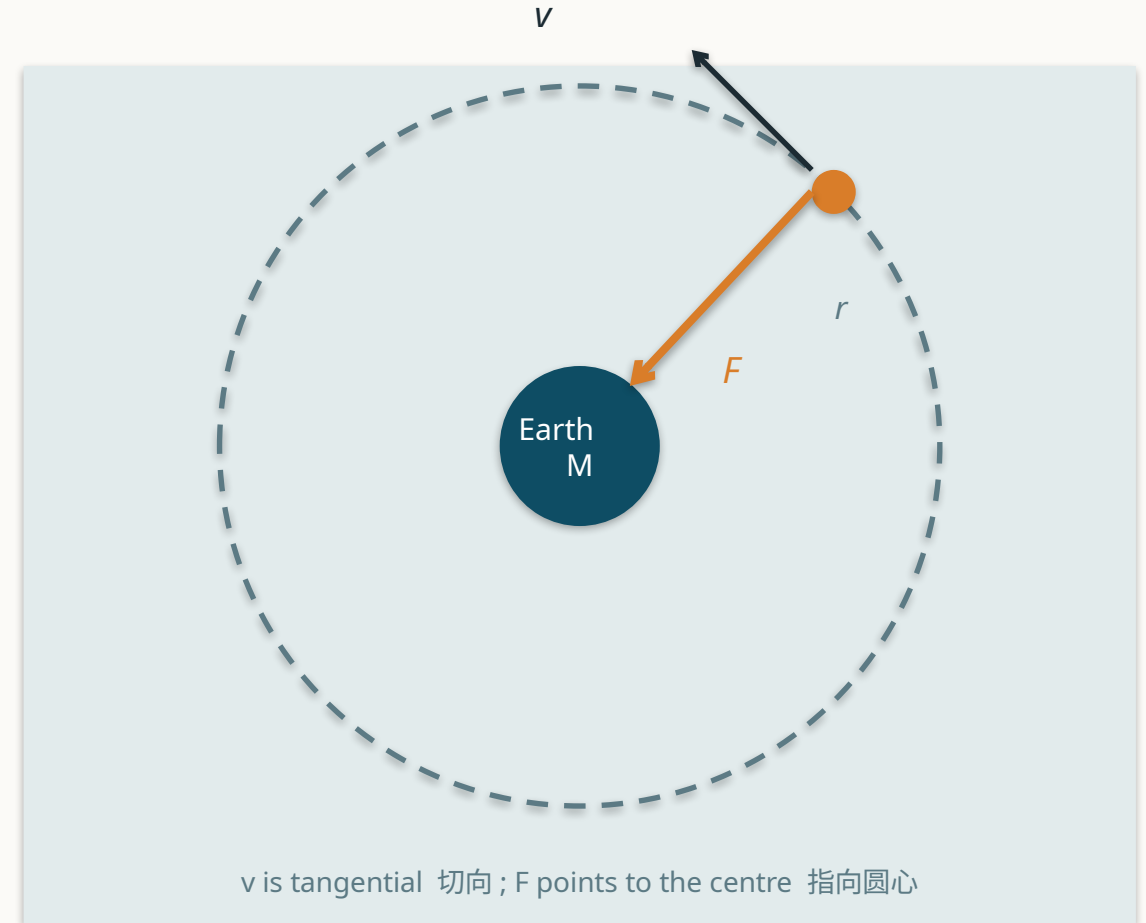
orbital speed 轨道速度
 m cancels — mass of satellite irrelevant

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$T^2 \propto r^3$ — Kepler's third law
 开普勒第三定律

- For the same central body, T^2/r^3 is constant 同一中心天体 T^2/r^3 为定值
- Closer orbits are faster and shorter-period 轨道越低，速度越大、周期越短
- The satellite is continuously accelerating — not in

equilibrium 卫星持续加速，并非平衡态



The Geostationary Orbit

地球同步轨道 — 通信卫星的 "定点" 轨道

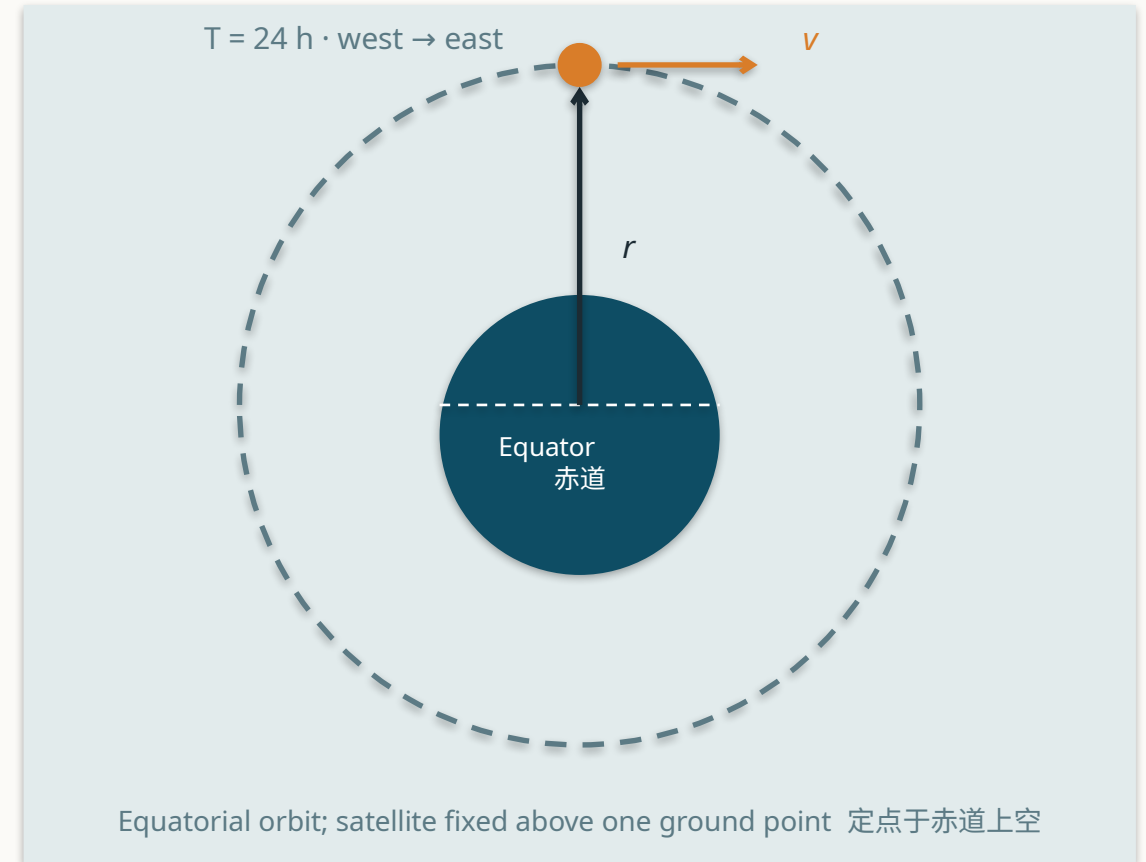
A **geostationary satellite** remains above the same point on the Earth's surface. It must have:

1. period **$T = 24 \text{ h}$** , same as the Earth's rotation 周期与地球自转相同
2. orbit **directly above the Equator** 轨道在赤道正上方
3. motion **west to east**, same direction as the Earth 自西向东

Key values 关键数据 (from $T^2 = 4\pi^2 r^3 / GM$):

from the centre; height ;

Uses 应用 : communication and TV relay satellites — ground dishes can point in a fixed direction 天线可固定指向 .



Gravitational Potential

引力势 — 从无穷远移动单位质量所做的功

The **gravitational potential ϕ** at a point is the work done **per unit mass** in bringing a small test mass from infinity to that point. Unit: J kg^{-1} .

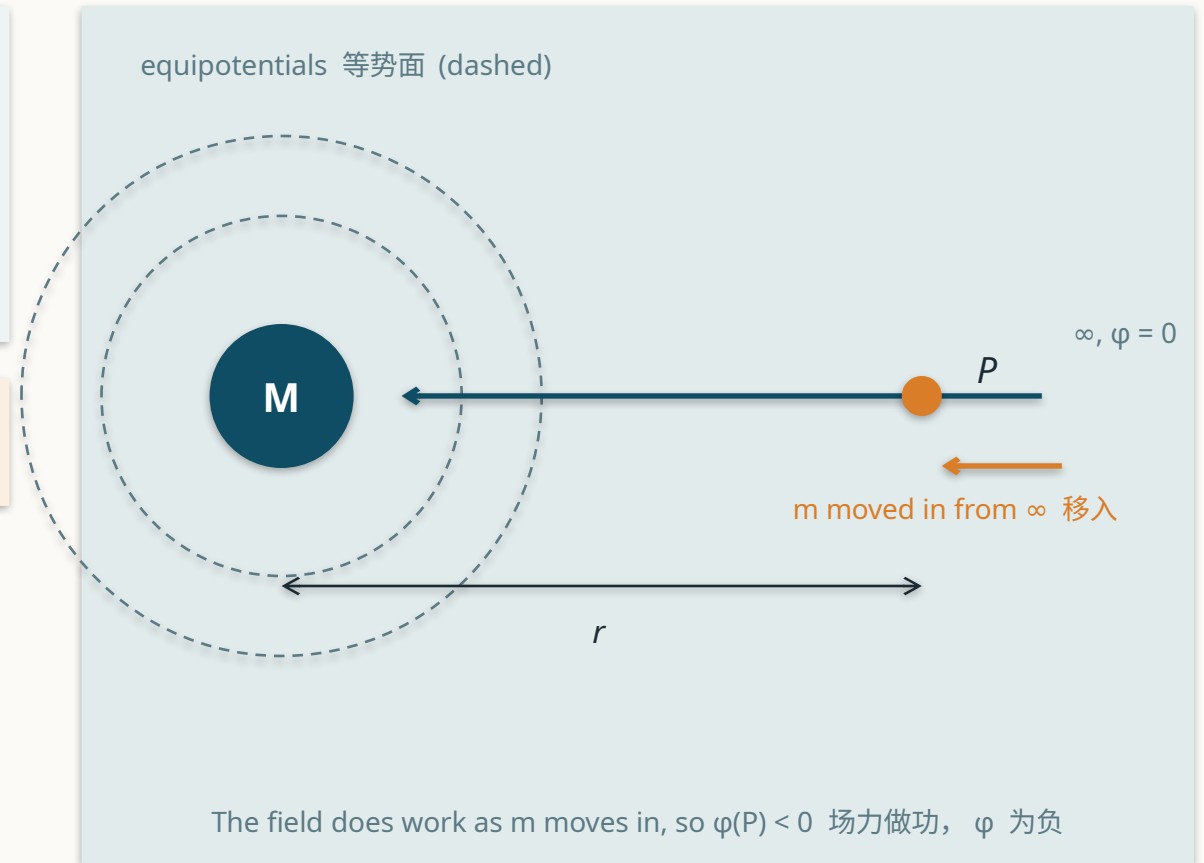
引力势：把单位质量从无穷远移到该点所做的功。

$$\phi = -\frac{GM}{r}$$

zero of potential at infinity
无穷远处 $\phi = 0$

- ϕ is **negative**: the field does the work, so energy is released as the mass falls in 场力做功, ϕ 为负
- ϕ increases with r , approaching 0 at infinity 随 r 增大而趋近于零
- Work to move a mass m : 移动质量 m 所需功
- Equipotential surfaces are concentric spheres 等势面为同心

球面



Potential Energy & Escape Velocity

引力势能与逃逸速度 — 从地表挣脱引力束缚所需的最小速度

$$E_p = m \varphi = - \frac{GMm}{r}$$

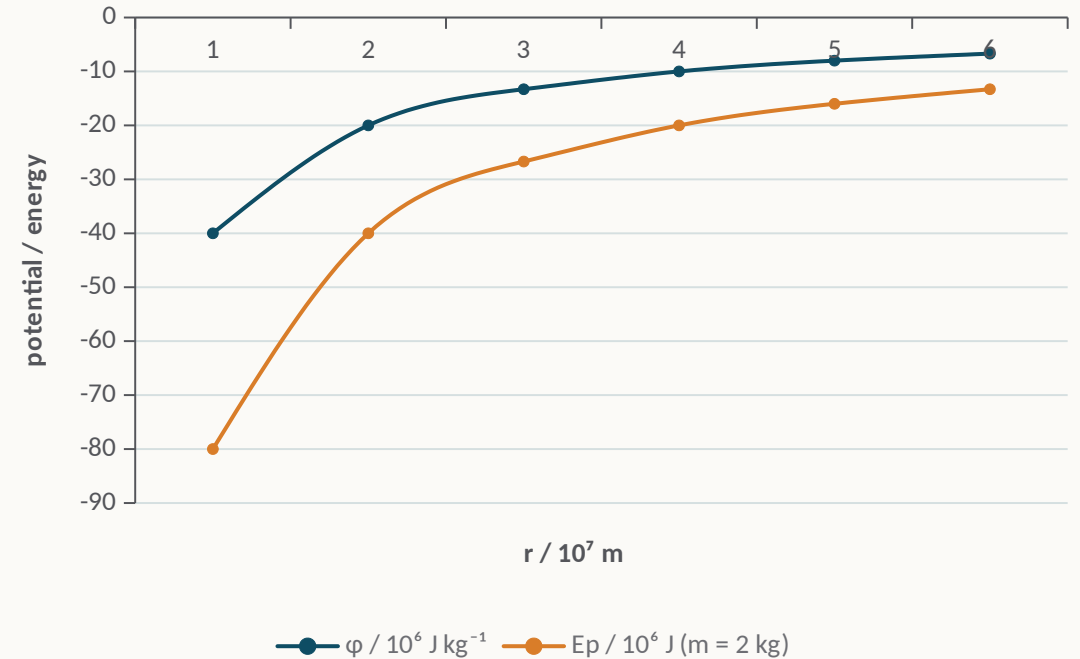
m = test mass 质量
 r = centre distance 质心距

Deriving escape velocity 推导逃逸速度

1. Work to move m from the surface ($r = R$) to infinity:
2. Kinetic energy supplies this work:
3. Mass m cancels — escape speed is independent of the object's mass 与物体质量无关

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

Earth 地球:
 $v \approx 11.2 \text{ km s}^{-1}$



Both φ and E_p are negative and rise towards 0 as $r \rightarrow \infty$ 均为负值并随 r 增大趋于零

Worked Example 13A

例题解析

WORKED EXAMPLE 13A

The masses of the Sun and the Earth are m_s and m_e . The separation of their centres is r .

Calculate the force of attraction between the Sun and the Earth.

太阳与地球间的万有引力计算；注意先把 km 换算成 m 。

SOLUTION 解答

- Step 1** Convert units: .
Step 2 Apply Newton's law:
Step 3 Evaluate:

Answer

Worked Example 13B

例题解析

WORKED EXAMPLE 13B

For a geostationary satellite, calculate: (a) the height above the Earth's surface; (b) the speed in orbit.
(Radius of Earth ; mass of Earth ; period .)

同步卫星高度与速度； r 是到地心的距离，高度要减去地球半径。

SOLUTION 解答

Step 1 Equate gravity to centripetal force: with gives .

Step 2 , so .

Step 3 Height ; speed .

Answer (a) (b)

Worked Examples 13C & 13D

例题解析

WORKED EXAMPLE 13C

Earth radius , surface field . (a) Find the mass of the Earth. (b) Find at the Moon's orbit .

- 1.
- 2.

Answer (a) (b)

WORKED EXAMPLE 13D

How much work is done by the gravitational field in moving a mass from infinity to a point from a mass ?

- 1.
2. Field does positive work:

Answer

Class Practice

随堂练习

Q1

Pluto has mass m_P and the Sun has mass M_S . Calculate the force of attraction between them when their separation is r .

Q2

A meteorite of mass 220 kg falls from rest at infinity to the Earth's surface (R_E). Calculate (a) its change in gravitational potential energy; (b) the speed at which it strikes the Earth.

Q3

Explain why gravitational potential is negative everywhere, and what the zero of potential is. 为什么引力势处处为负?

Answers Q1 Q2 (a) (b) Q3 potential is defined zero at infinity; the field does positive work pulling mass in, so potential decreases below zero.

Summary

本章小结

KEY FORMULAS 核心公式

$$g = \frac{F}{m}$$

field strength 场强定义

$$F = \frac{G m_1 m_2}{r^2}$$

Newton's law 万有引力

$$g = \frac{G M}{r^2}$$

point mass 点质量场强

$$v = \sqrt{\frac{G M}{r}}$$

orbital speed 轨道速度

$$T^2 = \frac{4 \pi^2 r^3}{G M}$$

orbital period 轨道周期

$$\phi = - \frac{G M}{r}$$

potential 引力势

$$E_p = - \frac{G M m}{r}$$

potential energy 引力势能

$$v_{esc} = \sqrt{\frac{2 G M}{R}}$$

escape velocity 逃逸速度

KEY IDEAS 核心思想

- A gravitational field is a region where a mass experiences a force; field strength is force per unit mass. 场强 = 单位质量受力。
- Gravity provides the centripetal force for orbits — this links Topics 12 & 13. 引力充当向心力是轨道问题的核心。
- Potential and potential energy are negative because zero is set at infinity. 无穷远为零点 $\Rightarrow$ 势能处处为负。

Exam Tips

备考提示

- **Use the centre-to-centre distance r** , not the height above the surface — the most common loss of marks. 用质心间距而非高度。
- State that **gravity provides the centripetal force** before writing — this earns the first mark. 先写 "引力提供向心力"。
- **Negative sign:** ϕ and V are negative; work done *by the field* bringing mass in from infinity is positive. 分清 "场做功" 与 "外力做功" 符号。
- Convert all units to SI: km $\rightarrow$ m, hours $\rightarrow$ s. Quote g from the data sheet. 统一 SI 单位。
- Geostationary answers need **all four conditions**: 24 h period, above the Equator, west to east, same point above ground. 同步轨道条件要写全。